

Engineering Physics (2025)
Course code 25PY101
Module 2 Unit 2: Optoelectronics

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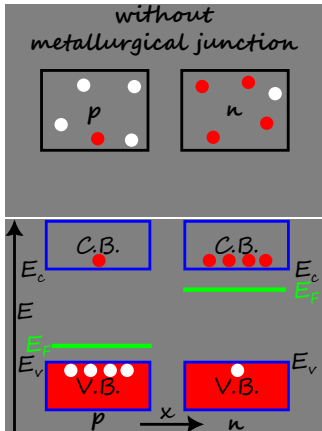
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- 1 p-n junction diode – forward and reverse conditions

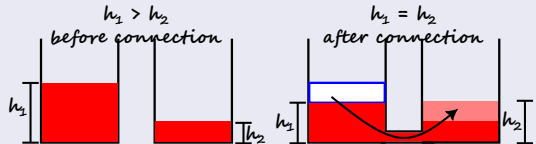
- 1 p-n junction diode – forward and reverse conditions

Isolated p-type and n-type semiconductors

- The Fermi level of p-type semiconductor is closer to the valence band edge E_v .
- The Fermi level of n-type semiconductor is closer to the conduction band edge E_c .



Analogy of gravitational force

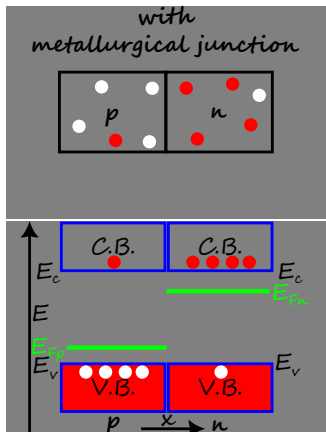


- When the glasses are connected, the gravitational force acts as driving force to reach equilibrium.
- In the case of semiconductors, the driving force to reach equilibrium is E_F .

pn junction – Metallurgical junction

Definition

Metallurgical junction is defined as the interface between the p-type and n-type semiconductors.

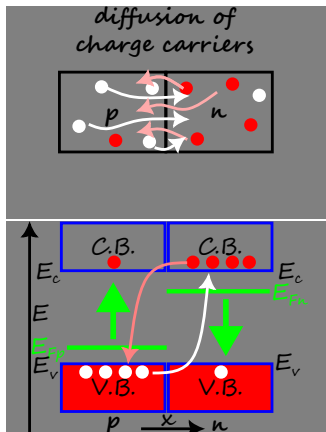


- Upon the formation of metallurgical junction, there is a difference in the Fermi level in both the regions.
- The n region has higher Fermi level E_{Fn} than p region E_{Fp} because there are more electrons in the n-region.
- The difference in Fermi level acts as a “driving force” so that electrons from n-region **diffuse** into the p-region.
- By similar analysis, holes from p-region diffuse into the n-region.

pn junction – Diffusion

Definition

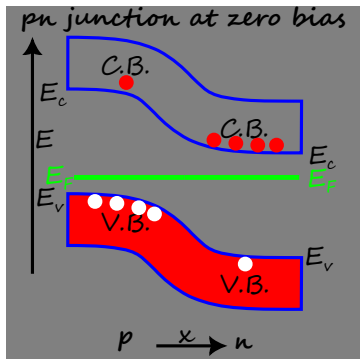
Diffusion is the net movement of particles from a region of higher concentration to a region of lower concentration.



- Due to the diffusion of mobile charge carriers, their ionic counterparts are left without partners.
- Diffusion of electrons from n-region lowers its Fermi level E_{Fn} .
- Diffusion of holes from p-region raises its Fermi level E_{Fp} .

pn junction at zero bias

pn junction at zero bias is at thermal equilibrium



Definition

Thermal equilibrium occurs when the net forces on the system are zero.

- Since the difference in Fermi level is a “driving” force, when the system reaches thermal equilibrium, the net force is zero.
- This implies the Fermi level is a constant throughout the system at thermal equilibrium.
- Since the external force is also zero, the pn junction at thermal equilibrium is said to be at zero bias ¹.

Key Insight

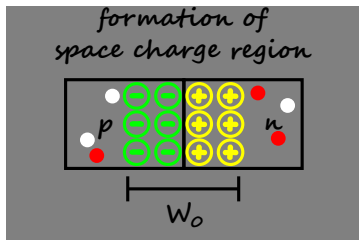
At thermal equilibrium, E_F is a constant.

¹The word bias is used in the context of electricity to refer to external voltage.

pn junction at zero bias – Space-charge region

Definition

- **Space charge region**, also called the **depletion region**, is the region created as a result of diffusion of free charge carriers across the metallurgical junction.
- The width of the depletion region is called depletion region width W_0 .



- Since the space charges consist of dopant ions, they are immobile.
- Space charge region is also called the **depletion region** since the region is depleted^a of the free charge carriers.

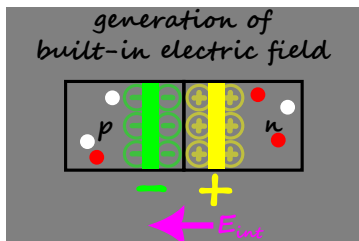
^adeplete means decrease or remove.

- The depletion region consists of positively charged donor ions on the n-side of the junction and negatively charged acceptor ions on the p-side of the junction.

pn junction at zero bias – Built-in electric field

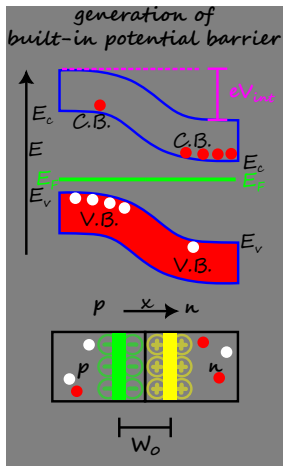
Definition

Built-in electrical field, also called internal electric field E_{int} is the electric field generated due to the oppositely charged regions on either side of the metallurgical junction.



- Within the depletion region, the oppositely charged ions on either side of junction can be considered as plates of a capacitor.
 - Since the plates are charged, a **built-in electric field** is generated.
 - The built-in electric field is directed from **n-region to p-region**.
- E_{int} prevents further build up of space charge due to diffusion as it opposes the diffusion of electrons from n-region into p-region and vice versa.

pn junction at zero bias – Built-in potential barrier



Definition

Built-in potential barrier V_{int} is the potential difference generated across the oppositely charged halves of the depletion region.

- If we model the depletion region as a capacitor with electric field E_{int} between the plates and the separation between the plates is W , then the potential difference across the plates is given by

$$E_{int} = \frac{V_{int}}{W_0} \Rightarrow V_{int} = E_{int} W_0$$

Key Insight

Potential barrier is proportional to the built-in electric field.

pn junction at non-zero bias

pn junction – at non-zero bias

- At non-zero voltage bias, there is an application of external force on the charge carriers. As a result, the pn junction cannot attain thermal equilibrium.
- Since there is no thermal equilibrium, the Fermi energy is not a constant across the pn junction.
- Since the direction of built-in electric field is from n-region to p-region, the behaviour of p-n junction is different when the external field E_{ext} is parallel or anti-parallel to the built-in electric field E_{int} .
 - 1 $E_{\text{ext}} \uparrow \downarrow E_{\text{int}}$ corresponds to forward biasing.
 - 2 $E_{\text{ext}} \uparrow \uparrow E_{\text{int}}$ corresponds to reverse biasing.

Key Insight

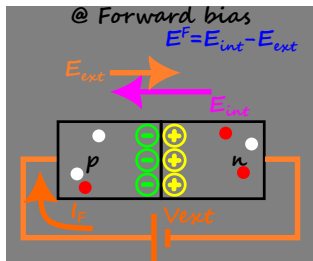


At non zero bias, the Fermi energy is not a constant across the pn junction.

pn junction – at forward bias

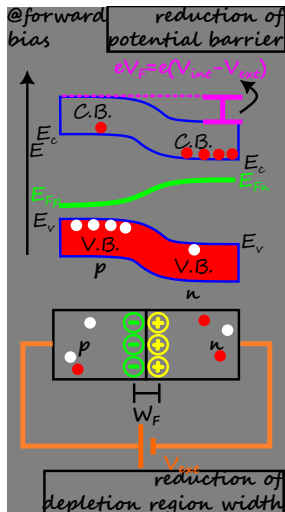
Definition

Forward bias condition is considered when the positive electrode of battery is connected to p-region and negative electrode is connected to n-region.



- The external field E_{ext} is from p-region to n-region.
- The internal field E_{int} is **anti-parallel** to E_{ext} .
- The net electric field is lower than the case of zero bias.
- So less space charge is created in depletion region. So we have smaller space charge width.
- Since potential is proportional to electric field, potential barrier is lowered.

pn junction – at forward bias



- From the band diagram, Fermi energy of n-region is higher than Fermi energy of p-region.
- So electrons will move from n region to p-region. This leads to current from p-region to n-region. This is called forward bias current I_F .
- Since electrons are present in n-region this leads to **good conduction** of pn-junction.

Compare with zero bias

$$E^F < E_{int}$$

$$V_F < V_{int}$$

$$E_{Fn} - E_{Fp} > 0$$

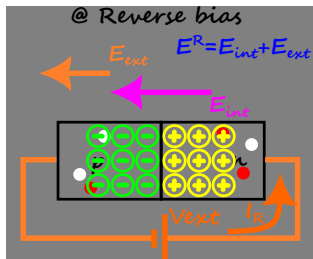
$$W_F < W_0$$

$$I_F \text{ flows } p \rightarrow n$$

pn junction – at reverse bias

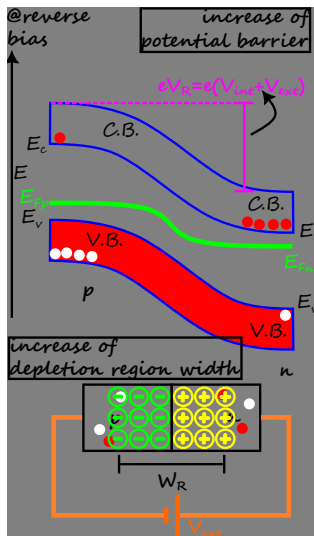
Definition

Reverse bias condition is considered when the positive electrode of battery is connected to n-region and negative electrode is connected to p-region.



- The external field is from n-region to p-region.
- Under reverse bias condition, the internal field is **parallel** to the external field.
- The net electric field is higher than the case of zero bias.
- So more space charge is created in depletion region. So we have larger space charge width.
- Since potential is proportional to electric field, potential barrier is raised.

pn junction – at reverse bias



- From the band diagram, Fermi energy of n-region is lower than Fermi energy of p-region.
- So electrons want to move from p region to n-region. But only minority carriers are there in p region.
- Since very few electrons are present in n-region this leads to **poor conduction** of pn-junction and the current is called reverse bias current I_R .

Compare with zero bias

$$E^R > E_{int}$$

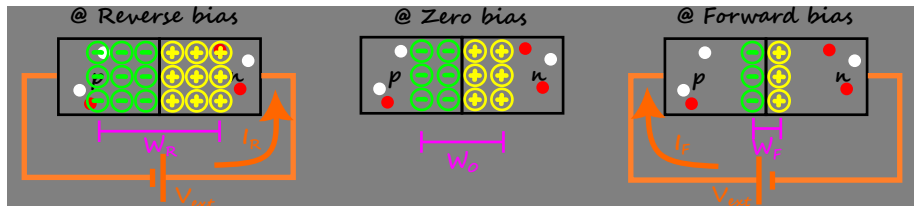
$$V_R > V_{int}$$

$$E_{Fn} - E_{Fp} < 0$$

$$W_R > W_0$$

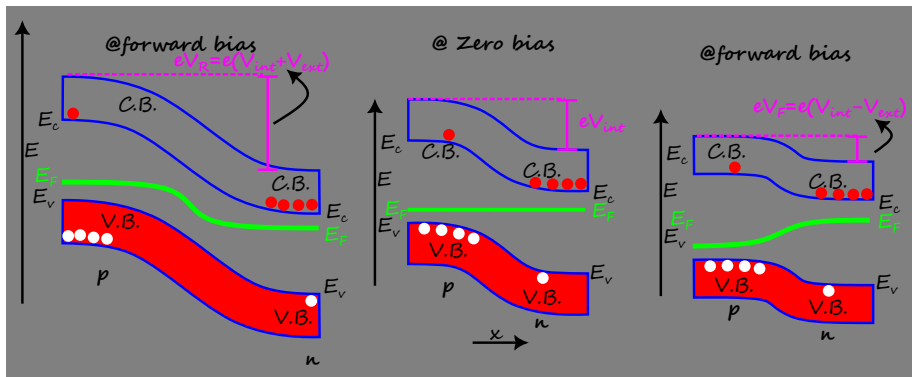
$$I_R \text{ flows } n \rightarrow p$$

Comparison of depletion region widths at different biases



$$W_R > W_0 > W_F$$

Comparison of potential barrier at different biases

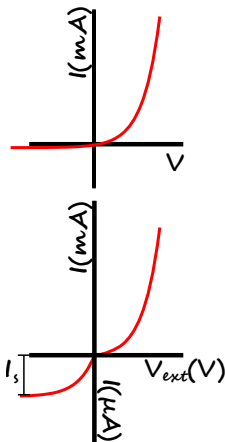


$$V_R > V_{int} > V_F$$

pn junction – $I - V$ characteristics

Definition

Rectifying action of pn junction is its property to allow conduction **unidirectionally**.



- The diode equation can be shown to be

$$I = I_s \left[\exp \left(\frac{eV}{k_B T} \right) - 1 \right]$$

where I_s is the reverse saturation current.

- The equation and the $I - V$ capture the non-linear nature of p-n junction.
- In the forward bias condition, it acts as a “closed switch” allowing large current to flow through it.
- In the reverse bias condition, it acts as an “open switch” allowing insignificantly low current to flow through it.
- This unidirectional conduction is called rectifying action.

pn junction – Reverse Saturation Current

Definition

The drift current due to minority carriers is known as **reverse saturation current**.

- As the reverse bias increases in magnitude, the reverse bias current saturates to a constant and is given by

$$\begin{aligned} & \lim_{V \rightarrow -\infty} I_s \left[\exp \left(\frac{eV}{k_B T} \right) - 1 \right] \\ &= I_s [\exp(-\infty) - 1] \\ &= I_s(0 - 1) \\ &= -I_s \end{aligned}$$

- This saturation current in the reverse bias direction is called reverse saturation current.
- Since the minority carrier concentration does not depend on the bias and only depends on the temperature, it is a constant at a given temperature.

Conceptual Questions

Problem

What causes majority carriers to flow at the moment when p-region and n-region are brought together? Why does this flow not continue until all carriers have recombined? [Hint for second part: Convince yourself that the built-in electric field opposes the diffusion “forces” for both the electrons and holes.]